Personal Learning Journey Report

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Role: Data Preprocessing and Feature Engineering

1. My Contribution and Responsibilities

My primary responsibility was to turn SQL data into a clean mathematical format that would fit our machine learning models. The original dataset had raw and post-match variables. To stop data leakage, I carefully left out all in game statistics such as corners, cards, shots and possession because those numbers only appear during or, after the match. Instead, I Built a sturdy set of pre-match features. This work included recording team strength by computing rolling form looking at goals scored and conceded over the last five games calculating rolling goal differences and adding pre-match bookmaker odds (B365H, B365D, B365A) as early probability hints. This careful feature selection made sure that the model used information that exists before kickoff, keeping the project firmly focused on our business goal of predictive strategic evaluation.

2. Feature Engineering: The 5-Match Rolling Form

Team names by themselves do not have any mathematical value. So, I created six features to show how teams perform. These features include rolling averages of goals scored goals allowed and goal differences for both home teams and away teams.

Why 5 matches? I chose a 5-match window because that is what is commonly used in sports analytics. It gives a balance between showing how a team is doing right now, which is about a month of football, and removing the noise from one lucky win or old data from matches played several months ago when things might have been different.

Leakage Prevention: To do this I used Pandas.rolling() along with.shift(1). The.shift(1) was very important. It made sure that the goals from the match were not part of the historical average. This helped stop data from getting into the model while it was learning.

3. Handling Outliers & Avoiding PCA

Based on Member 1's EDA, we observed outliers such as 9-10 goals in certain matches. However, because these are valid football results and not data entry errors, I chose not to drop these rows. Instead, I handled them by scaling the data. Furthermore, I deliberately avoided using Principal Component Analysis. Our business stakeholders require interpretable. PCA would have scrambled our engineered form features into uninterpretable components, ruining the business value of our model.

4. Technical Challenges

1. **Cold Start Problem:** Missing historical data caused NaN values for early season matches and newly promoted teams.

2. **Label Encoding Bias:** Numerical labels for teams and leagues could introduce false relationships between categories.

3. **Changing Team Performance:** Transfers, injuries, and manager changes can make historical forms less reliable for current predictions.